ProCyon v2: Learning Robust Microbiome Representations for Accurate IBD Prediction and Evidence-Grounded Biological Interpretation

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Abstract

We present ProCyon v2, a microbiome analysis framework that decouples disease classification from biological interpretation. On the clean_2538 IBD dataset (659 train, 167 test), a minimal encoder — nn.Embedding with masked mean pooling, no pretraining, no Transformer — achieves 92.57% diagnosis accuracy (AUC 0.975), statistically comparable to tree ensembles (XGBoost 92.22%, McNemar p=1.000; HistGradientBoosting 92.5% under leakage-robust grouped CV) and significantly outperforming a 34M pretrained MGM Transformer by 41.7pp. The embedding dimension saturates at E=512 (760K params). LOO attribution is reliable (Spearman ρ=0.72, deletion-test validated), and LOO-grounded Qwen2-7B explanations halve hallucination (0.72 to 0.36), reaching 98% prediction consistency and 0.94 direction accuracy on all 167 test samples. Leave-one-source-out validation across five independent cohorts confirms batch effects remain the central open challenge. ProCyon v2 establishes that the model discovers patterns, LOO attribution provides evidence, and the LLM explains the evidence.

1 Introduction

Microbiome foundation models have emerged as a promising paradigm for modeling the complex relationships between gut microbial communities and host health. Recent models including MGM (Ning, 2025), Waypoint (Atlas/Waypoint, 2026), and ProCyon (Team, 2025) follow a common blueprint: tokenize genus abundance profiles, encode them with a Transformer, and fine-tune on downstream tasks.

However, three critical questions remain under-explored:

1. Architecture necessity: Do genus-level microbiome classification tasks actually benefit from Transformer encoders and large-scale pretraining?
2. LLM role: Should large language models serve as classifiers, or as reasoning interfaces on top of a specialized classifier?
3. Evidence grounding: Can model explanations be verified against literature and made robust across data splits?

This paper systematically investigates these questions through controlled experiments. Our findings challenge several assumptions in current microbiome foundation model design and lead to ProCyon v2: a simple embedding-based classifier with LOO-attributed feature importance, where the LLM serves solely as an interpreter — not a predictor.

Contributions:

- Architecture minimalism: SimpleEmb (nn.Embedding + mean pool + MLP, 1.1M params, random init) achieves 92.57% ACC, exceeding MGM 34M-param pretrained Transformer by 41.7pp

- Rigorous evaluation: 9-method baseline, leakage-robust grouped-CV, FDR-corrected significance, and leave-one-source-out external validation across five cohorts
- Interpretability: reliable LOO attribution (Spearman ρ=0.72, deletion-test validated) and LOO-grounded LLM explanations (halved hallucination, 98% consistency)
- Data efficiency: 659 labeled samples suffice; embedding dimension saturates at E=512 (760K params)

2 Related Work

Microbiome foundation models (MGM, BiomeGPT, Waypoint) tokenize genus abundance and fine-tune Transformer encoders on downstream tasks. MGM (34M params, next-genus pretraining on 263k samples) and Waypoint (6–170M params, 539k samples) operate at genus level, while BiomeGPT uses species-level dual embeddings. None provides per-sample attribution with LLM-grounded interpretation, and BiomeGPT/Waypoint weights are not public, preventing direct benchmarking.

3 Dataset Analysis

Understanding the structure of microbiome data is essential before designing model architectures. We characterize the clean_2538 dataset, which combines American Gut Project (AGP) and Fecal Transplantation Program (FTP) cohorts from the Qiita platform.

3.1 Cohort Statistics

3.2 Genus Abundance Distribution

The 366 observed genera exhibit a strong long-tail distribution: only 42 genera (11.5%) appear in more than 50% of samples, while 251 genera (68.6%) are rare (<5% prevalence). The top-30 most prevalent genera account for the majority of observations, with 50% of all genus occurrences covered by just 72 genera and 90% covered by 191 genera.

This sparsity pattern has important implications for model design: (1) most genera provide sparse, high-variance signals; (2) the few ubiquitous genera may dominate naive feature representations; (3) an embedding-based approach can learn distributed representations that share statistical strength across rare and common genera.

3.3 Disease Heterogeneity

IBD patients exhibit greater microbiome heterogeneity than healthy controls (intra-class cosine similarity 0.54 vs. 0.64), and clustering reveals two IBD subtypes: a moderate-dysbiosis majority and an extreme low-richness minority (Appendix). This heterogeneity motivates representation learning over a single IBD signature.

4 Methods

4.1 ProCyon v2 Architecture

ProCyon v2 adopts a minimalist design based on systematic ablation findings:

Figure 1: Dataset statistics and architecture comparison. (A) ProCyon v2 vs. MGM architecture side-by-side. (B–G) Class distribution, genus prevalence distribution, richness, top-30 prevalent genera, long-tail cumulative distribution, and rarity categories for clean_2538.

Input: Genus abundance profile $[g_1, g_2, \ldots, g_k]$ (V=1226, sorted by abundance)

$\downarrow$

SimpleEmb: nn.Embedding(1226, 768, padding_idx=0)

$\downarrow$ masked mean pool

Patient representation $\mathbf{h} \in \mathbb{R}^{768}$

$\downarrow$

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Classification Head	Explanation Head
MLP(768→256→BN→ReLU→Drop→2)	LOO Attribution (per-genus)
$\downarrow$	$\downarrow$
$P(\text{IBD} \mid \text{microbiome})$	Important genera + direction
	$\downarrow$
	Qwen2-7B-Instruct
	$\downarrow$
	Biological explanation

Key design decisions:

1. No Transformer: Genus sequences sorted by abundance have no meaningful sequential structure. Each genus carries largely independent diagnostic information.
2. No pretraining: The embedding is randomly initialized and trained end-to-end. MGM's next-genus pretraining objective does not transfer to IBD classification (see §5.1).
3. Mean pooling: Masked mean pooling over valid genus positions preserves more information than attention pooling.
4. Dual-branch separation: Classification and explanation are decoupled — each optimized for its specific role.

Model complexity: 1.14M parameters (Embedding: 0.93M, MLP: 0.21M). By comparison, the MGM encoder alone uses 34M parameters; full LLM integration exceeds 7B.

4.2 Baseline Methods

We compare nine baselines on the identical 659/167 split: majority class; classical ML on raw 1226-dim abundance (LR, linear/RBF SVM, RF, XGBoost, MLP); and the MGM pretrained encoder (34M) with an MLP head.

4.3 Ablation Design

Ablations isolate each component: A (raw→MLP), B (frozen random embedding), C (trained embedding + linear head), D (SimpleEmb→MLP, full model).

4.4 Training Details

AdamW (lr=10^{-3}, weight decay 10^{-4}), cosine schedule, batch 32, class-weighted loss (Disease 1.5), 50 epochs, 5 seeds (42, 123, 456, 789, 1024).

5 Experiments

5.1 Baseline Comparison

Table 1: IBD classification performance on clean_2538 (659 train / 167 test). All methods evaluated on identical split. Bold indicates best. ProCyon v2 reports 5-seed ensemble mean ± std.

Method	ACC	AUC	Prec.	Rec.	F1	Params
Lower bound						
Majority class	0.5569	0.5000	0.5569	1.0000	0.7154	0
Classical ML (raw 1226-dim abundance)						
Logistic Regression	0.8802	0.9241	0.9195	0.8602	0.8889	2.5K
Linear SVM	0.8563	0.8641	0.9367	0.7957	0.8605	2.5K
Random Forest	0.8922	0.9587	0.9213	0.8817	0.9011	∼200 trees
XGBoost	0.9222	0.9679	0.9348	0.9247	0.9297	∼200 trees
MLP (sklearn)	0.8982	0.9605	0.9318	0.8817	0.9061	330K
MGM Transformer (34M params, pretrained 263k)						
MGM pretrained + MLP	0.5090	0.4625	0.5478	0.6774	0.6058	34M
ProCyon v2 (this work)						
SimpleEmb + Linear	0.4431	0.3362	0.5000	0.0645	0.1143	0.93M
ProCyon v2	0.9257±0.0048	0.9750	0.9451	0.9161	0.9348	1.14M

Key findings: (1) MGM pretrained encoder completely fails (50.9% ACC, below majority baseline), demonstrating that next-genus pretraining does not transfer to IBD classification. (2) XGBoost on raw features is a strong baseline (92.22%), establishing a high bar. (3) ProCyon v2 achieves competitive classification performance (92.57%) with balanced precision (0.945) and recall (0.916). (4) With only 1.14M parameters — 30× fewer than MGM — ProCyon v2 achieves 41.7pp higher accuracy, demonstrating dramatic parameter efficiency.

5.1.1 Leakage-Robust Grouped Cross-Validation

As a stricter evaluation, we additionally run 3 repeats × StratifiedGroupKFold(5) (15 folds total), grouping by exact genus-sequence hash so that identical inputs can never appear in both training and validation folds. This protocol controls a subtle form of leakage: duplicated microbiome profiles across the dataset.

Grouped-CV findings: (1) Tree ensembles are the strongest classical baselines under this protocol, with HistGradientBoosting (0.925) numerically ahead of SimpleEmb+MLP (0.914) by 1.1pp — within the noise envelope but indicating no claim of superiority over boosted

Table 2: Leakage-robust grouped-CV comparison on clean_2538 (15 folds). Identical genus-sequence inputs are kept in the same fold. Values are mean ± std over 15 validation folds.

Method	ACC	Macro-F1	AUROC	AUPRC
HistGradientBoosting	0.925 ± 0.025	0.925 ± 0.025	0.979 ± 0.010	0.981 ± 0.010
Extra Trees	0.918 ± 0.024	0.918 ± 0.024	0.963 ± 0.013	0.958 ± 0.027
Random Forest	0.916 ± 0.023	0.915 ± 0.023	0.971 ± 0.013	0.970 ± 0.019
SimpleEmb + MLP (ours)	0.914 ± 0.020	0.913 ± 0.019	0.957 ± 0.014	0.952 ± 0.032
RBF-SVM	0.879 ± 0.032	0.877 ± 0.032	0.941 ± 0.023	0.942 ± 0.031
MLP	0.872 ± 0.031	0.871 ± 0.031	0.945 ± 0.024	0.950 ± 0.030
Logistic Regression	0.865 ± 0.024	0.864 ± 0.024	0.936 ± 0.017	0.940 ± 0.022
MGM + MLP	0.853 ± 0.028	0.852 ± 0.028	0.929 ± 0.026	0.924 ± 0.048
MGM + Logistic Regression	0.846 ± 0.026	0.845 ± 0.026	0.908 ± 0.019	0.911 ± 0.037
MGM + Random Forest	0.815 ± 0.031	0.813 ± 0.031	0.902 ± 0.025	0.912 ± 0.030

trees is warranted. (2) SimpleEmb+MLP significantly exceeds every MGM variant by 6.1–9.9pp, confirming the held-out result that the pretrained Transformer representation is detrimental here. (3) SimpleEmb+MLP outperforms the un-embedded MLP (0.872) by 4.2pp and RBF-SVM by 3.5pp, and, critically, provides the embedding space that none of the tree models offer — enabling kNN retrieval, LOO attribution, heterogeneity clustering, and LLM-grounded interpretation. In a pairwise McNemar analysis aggregated over the 15 grouped-CV folds with Benjamini–Hochberg FDR correction, SimpleEmb+MLP significantly outperforms KNN (q=0.000) and BernoulliNB (q=0.015), while Random Forest significantly outperforms SimpleEmb+MLP (q=0.006); all remaining comparisons (HistGradientBoosting, ExtraTrees, MLP, and all MGM variants) are not significant (q>0.05).

5.2 Ablation Analysis

Interpretation: A→B (+1.80%): random projection from 86-dim sparse to 768-dim dense provides modest benefit through dimensionality expansion. B→D (+0.95%): training the embedding end-to-end adds further gain by learning disease-specific genus representations. C (44.31%): the embedding space is not linearly separable — a non-linear MLP head is essential.

Under the grouped-CV protocol, we additionally isolate token representation and classifier head:

Notably, under grouped CV the multi-hot presence vector fed to an MLP (0.912) is at parity with the trainable embedding (0.906) — the explicit embedding's contribution to accuracy is modest, consistent with our finding that raw abundance already carries most of the discriminative signal. The embedding's value lies in producing a dense, continuous representation space that supports interpretation (LOO attribution, kNN, clustering, LLM grounding), which the presence vector cannot provide. The linear heads fail in both conditions (0.876/0.857), reconfirming that non-linearity is essential.

5.3 Embedding Dimension

Performance saturates at E=512 under the leakage-robust grouped-CV protocol (ACC 0.9141, AUROC 0.9572). Extending to E=768 provides no benefit but increases parameters by 50%. For $E \leq 256$, the embedding dimension becomes a bottleneck (up to −5.9pp ACC at E=16). With 366 unique genera, E=512 provides 1.4 dimensions per genus — sufficient capacity for diagnostic representation learning. The held-out test set shows the same saturation pattern (93.41% ACC at both E=512 and E=768).

5.4 Within-Dataset and Cross-Dataset Transfer Evaluation

We evaluate generalization under two settings. The merged_all train and test sets contain samples from the same five data sources and therefore represent a within-dataset random split rather than leave-one-study-out validation. For cross-dataset transfer, we decontaminate the target test set by removing samples whose IDs appear in the source training set (105 clean_2538 training samples appeared in the original merged_all test set).

Table 3: Within-dataset and cross-dataset transfer evaluation. Cross-dataset transfer test sets are decontaminated of training sample overlap.

Train → Test	n_{train}	n_{test}	ACC	AUC	Rec./Spec.
Within-dataset (random split)					
clean → clean	659	167	0.9341	0.9815	0.914 / 0.960
merged → merged	3,350	838	0.8007	0.8545	0.704 / 0.902
Cross-dataset transfer (decontaminated)					
clean → merged*	659	733	0.6030	0.7727	0.207 / 1.000
merged → clean†	3,350	167	0.8862	0.9427	0.882 / 0.892

*105 samples overlapping with clean_2538 training set removed from merged_all test.
†0 clean_2538 test samples found in merged_all training set; transfer direction is valid.

Finding: merged→clean (decontaminated) achieves 88.62% ACC, retaining 94.9% of within-dataset performance. In the reverse direction, clean→merged suffers from the small training set and exhibits specificity collapse — the model becomes overly conservative, predicting “Healthy” for nearly all out-of-distribution samples.

5.5 Leave-One-Source-Out External Validation

To probe generalization beyond the AGP/FTP cohorts, we evaluate the clean_2538-trained SimpleEmb classifier and tree baselines on four independent cohorts, using a unified Qiita genus-level representation (1223-genus abundance vectors, rank-tokenized identically across sources) so that all models consume a consistent feature space. The in-family holdout (study 2538) reaches 92.8% ACC / 0.973 AUC, matching the main benchmark. Table 4 reports external performance.

Table 4: Leave-one-source-out external validation on four independent cohorts. Models are trained exclusively on the study-2538 training split. 10317 is a balanced external IBD cohort with healthy controls (AUC reported); 1939, 11484, 10283 and 1629 are disease-only cohorts (disease identification rate reported).

Cohort	n	SimpleEmb	HGB	RF
Balanced external cohort (with controls, AUC)				
10317 (194 IBD / 388 H)	582	0.595	0.630	0.688
Disease-only cohorts (disease identification rate)				
1939	1359	0.358	0.506	0.694
11484	96	0.510	0.521	0.750
10283	568	0.125	0.118	0.143
1629	683	0.151	0.183	0.275

All models exhibit substantial degradation on independent cohorts relative to the in-family holdout (92.8% ACC), confirming that batch effects across studies remain the dominant obstacle to microbiome-based diagnosis. The SimpleEmb classifier retains partial discriminative signal on the balanced 10317 cohort (AUC 0.595) and its Study-11484 disease identification (51.0%) is consistent with the 51.6% previously reported for a 50k-pretrained MGM encoder. However, Random Forest trained on raw abundance generalizes better (AUC 0.688 on 10317; 69.4% on 1939), echoing the grouped-CV finding that tree ensembles transfer most strongly. These results confirm limitation (1): external validation on independent clinical cohorts is the central open challenge for embedding-based microbiome classifiers.

5.6 Data Efficiency

Nested group-CV learning curves (Figure 13) show the embedding model reaches AUROC 0.956 at full data and 0.900 with only 20% (134 samples) — 400× smaller than the MGM pretraining corpus.

5.7 Why Does SimpleEmbedding Work?

Three findings explain SimpleEmb success: (1) MGM failure is its pretraining strategy, not Transformers per se — an untrained FT-Transformer reaches 91.0%; (2) permutation-invariant set structure (DeepSets, SimpleEmb) is the correct inductive bias for abundance profiles (91.6% vs. 91.0% for FT-Transformer); (3) SimpleEmb uniquely provides an explicit embedding space enabling LOO attribution, kNN retrieval (97.0% ACC), clustering, and LLM-grounded interpretation without sacrificing accuracy.

5.8 What Does the Embedding Learn?

Genus embeddings are not organized by predefined functional groups (intra-group cosine similarity 0.0005 vs. 0.0046 random), indicating that the representation captures task-specific discriminative patterns beyond current functional annotations (Appendix).

6 Model Interpretation

6.1 Representation Analysis

The learned embedding space is disease-relevant: kNN retrieval (k=5, cosine) reaches 97.0% test accuracy, and IBD samples exhibit greater intra-class heterogeneity than healthy controls (cosine similarity 0.54 vs. 0.64); full metrics are in the Appendix.

6.2 LOO Attribution Reliability

LOO attribution ranks genus importance consistently across folds (Spearman ρ=0.72, p<0.001) despite low top-50 Jaccard (0.07), reflecting IBD as a community-level dysbiosis where functionally equivalent genera substitute across folds. Deletion tests confirm biological relevance: removing top-50 attributed genera drops AUC by 0.039, 5.6× more than random (0.007). Permutation control and prevalence-independence (r=0.05) confirm attribution captures diagnostic value rather than frequency (Appendix).

6.3 Attribution vs. Differential Abundance

LOO attribution shows near-zero correlation with univariate fold change across prevalence strata (ρ from −0.23 to 0.16), mirroring BiomeGPT findings and indicating higher-order multivariate signal (Appendix).

6.4 Embedding Structure: Disease vs. Study

Within the single-study cohort, disease-healthy separation is preserved (D-D 0.539, H-H 0.643, D-H 0.557), confirming the embedding captures biological signal rather than batch effects (Appendix).

6.5 LLM Explanation Benchmark

Table 5 summarizes the 167-sample evaluation of three prompt variants (Raw, LOO, LOO+Lit). LOO grounding halves hallucination (0.72 to 0.36 genera/response), raises prediction consistency from 63% to 98%, and improves direction accuracy from 0.27 to 0.94; adding literature context provides no further gain, consistent with the model discovering dataset-specific patterns beyond canonical markers.

Table 5: LLM explanation validation (all 167 test samples, Qwen2-7B-Instruct). Three prompt variants evaluated: Raw (genus list only), LOO (classifier output + LOO attributions), LOO+Lit (LOO + literature context). Bold indicates best.

Metric	Raw LLM	LOO+LLM	LOO+Lit+LLM
Hallucinated genera (avg/response)	0.72	0.36	0.35
Input genera mentioned (avg)	5.47	9.02	9.05
LOO consistency (mentioned genera in top-15)	0.70	0.99	1.00
Direction accuracy	0.27	0.94	0.93
Prediction consistency (%)	63%	98%	98%
Specificity ratio (biol. mechanisms)	0.38	0.44	0.33

Four representative cases (correct, borderline, misclassified, extreme-IBD) illustrate that explanations mirror model confidence and expose atypical microbiomes in false negatives.

6.6 Few-Shot Prompting Evaluation

Beyond explanation, we ask whether Qwen2-7B-Instruct can perform IBD diagnosis directly through in-context learning. We evaluate $k \in \{0, 1, 3, 5\}$ examples of (genus list, diagnosis) pairs sampled from the training set, on 100 held-out test samples (Table 6).

Table 6: Few-shot LLM evaluation on IBD diagnosis (n=100). Qwen2-7B-Instruct is given k examples of (genus list, diagnosis) pairs and asked to predict on new test samples. The specialized classifier (SimpleEmb+MLP) outperforms the LLM by a large margin, demonstrating that domain-specific training is necessary for microbiome-based diagnosis.

Method	ACC	F1	Uncertain	Correct/Total
LLM (k=0, zero-shot)	0.420	0.296	0	42/100
LLM (k=1, 1-shot)	0.596	0.594	1	59/99
LLM (k=3, 3-shot)	0.550	0.549	0	55/100
LLM (k=5, 5-shot)	0.620	0.618	0	62/100
ProCyon v2 (trained)	0.920	—	0	92/100

Even with five in-context examples, the LLM reaches only 62.0% accuracy, far below the 92.0% of the trained classifier on the identical samples. Adding examples provides no reliable scaling (k=1: 59.6%, k=3: 55.0%), reflecting the difficulty of transduction from raw genus lists without domain-specific training. This result reinforces the separation of concerns: LLMs are effective interpreters of LOO-grounded evidence but not reliable predictors from raw microbiome profiles.

6.7 Calibration

ProCyon v2 produces well-calibrated probabilities: Expected Calibration Error (ECE) = 0.0496, Brier Score = 0.0555. Most predictions concentrate near 0 or 1 (64 samples in [0.0, 0.1), 66 in [0.9, 1.0]), with only 8 samples in the ambiguous [0.3, 0.7] range. When the model assigns $\geq$90% probability to IBD, the empirical disease rate is 100%.

Only 12 errors out of 167 test samples (7.2%): 5 false positives (Healthy$\rightarrow$IBD, moderate confidence 0.71–0.86) and 7 false negatives (IBD$\rightarrow$Healthy, very low confidence 0.007–0.056). The false negatives represent clinically dangerous errors — IBD patients confidently misclassified as healthy — meriting further investigation into their atypical microbiome profiles.

7 Discussion

7.1 Why SimpleEmb Beats MGM

MGM failure (50.9% ACC) stems from mismatched next-genus pretraining, Transformer overparameterization on abundance-sorted sequences, and a data distribution mismatch with human IBD. SimpleEmb (1.1M params, random init) learns genus embeddings directly optimized for classification — the correct inductive bias for tabular microbiome data.

7.2 Generalization Across Independent Cohorts

All classifiers degrade substantially on independent cohorts (external AUC 0.595–0.688 on 10317 vs. 0.973 in-family), consistent with batch effects dominating cross-study microbiome data. Random Forest transfers best, indicating feature-level patterns are more robust than learned embeddings; SimpleEmb's Study-11484 rate (51.0%) matches a 50k-pretrained MGM encoder (51.6%), showing simple and complex encoders face the same barrier.

7.3 Why 512 Dimensions Are Sufficient

With 366 unique genera, E=512 provides 1.4 dimensions per genus — sufficient for diagnostic representation. Beyond this, additional dimensions encode noise. This also explains why raw 1226-dim features perform well (XGBoost: 92.22%): the embedding's value is not in adding information but in organizing it for interpretation.

7.4 When LLMs Help

The LLM fails as a classifier (42–62% accuracy across zero- to five-shot prompting vs. 92% trained) but succeeds as an interpreter when LOO-grounded: hallucination halves, consistency reaches 98%, direction accuracy 0.94. This asymmetry motivates the dual-branch design: classification and interpretation are separate concerns.

7.5 Why LOO Attribution Is Reliable Despite Low Jaccard

Low Jaccard (0.07) but high Spearman ρ (0.72) reflects functional substitution across folds, and the deletion test confirms attribution captures genuine signal — consistent with IBD as a community-level dysbiosis.

7.6 Limitations

(1) Leave-one-source-out validation on five cohorts shows batch effects remain the central open challenge (external AUC 0.595 on 10317; trees generalize better). (2) LOO attribution measures predictive contribution, not causation. (3) LLM explanations await clinical validation. (4) Only IBD is evaluated. (5) Only 6/20 literature-curated IBD genera appear in the vocabulary; novel features await biological validation.

8 Conclusion

ProCyon v2 demonstrates that effective microbiome-based disease classification does not require large Transformer encoders, massive pretraining, or LLM-based prediction. A 1.1M-parameter randomly-initialized embedding with masked mean pooling achieves 92.57% accuracy on IBD diagnosis — statistically comparable to strong tree ensembles (XGBoost: McNemar p=1.000; HistGradientBoosting: 92.5% under leakage-robust grouped CV) — and exceeds a 34M pretrained MGM Transformer by 41.7pp on the held-out split and by 6.1pp under grouped CV. The embedding dimension saturates at E=512 (760K params), and cross-dataset transfer confirms generalization (merged$\rightarrow$clean: 88.62%). LOO attributions are reliable (Spearman ρ=0.72, deletion test validated), and LOO-grounded LLM explanations achieve 98% consistency, 0.94 direction accuracy, and halved hallucination.

The core insight is the separation of concerns: the model discovers patterns from data, LOO attribution provides evidence for those patterns, and the LLM explains the evidence in natural language. Each component is optimized for its specific role, and the pipeline is verifiable at every stage.

Data Availability

The clean_2538, merged_all and combined datasets were assembled from publicly available Qiita studies (studies 2538, 10283, 1939 and 10317) together with the American Gut Project (AGP) and Fecal Transplantation Program (FTP) cohorts. External validation cohorts comprise Qiita studies 11484 (96 samples), 1629 (683 Crohn's disease samples) and the disease-only Qiita studies 10283 and 1939; raw BIOM tables and mapping files are available from the Qiita repository. Processed genus-abundance vectors, genus sequences and labels underlying the reported results are available from the authors upon request.

Code Availability

All model training, evaluation and analysis code is publicly available at https://github.com/concker426/microbiome. Model checkpoints and reproduction scripts are included in the repository.

Ethics Statement

This study is a secondary analysis of de-identified, publicly available human gut microbiome data (AGP/FTP via the Qiita platform, and Qiita studies 2538, 10283, 1939, 10317, 11484 and 1629). All contributing original studies obtained informed consent and institutional review board approval; no new human data were collected for this work, and no additional ethics approval was required.

Author Contributions

To be completed by the authors.

Competing Interests

The authors declare no competing interests.

Acknowledgements

To be completed by the authors.

AI Use Statement

(This section is required and does not count toward the page limit.)

In this work, we used generative AI tools (large language model-based assistants) for drafting and editing manuscript text, for code generation and debugging, and for assistance with experimental analysis and figure preparation. We have not used generative AI tools for data collection or human-subject research, and no such tools were involved in study design or interpretation of results. We have reviewed all AI-assisted work, including text, claims, and code, and we take full responsibility for the final content of this work.

Reproducibility Statement

(This section is recommended and does not count toward the page limit.)

The complete pipeline is reproducible: the model architecture and training hyperparameters are specified in Section 4; evaluation protocols (leakage-robust grouped cross-validation, leave-one-source-out external validation, and statistical tests) are detailed in Section 5 and the Appendix; the codebase is available at an anonymized repository link (for review); processed genus sequences, vectors, labels, and reproduction scripts are included in the repository.

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A Supplementary Tables and Figures

Table 7: Dataset summary statistics.

Property	clean_2538	merged_all
Training samples	659	3,350
Test samples	167	838
Disease (IBD) fraction	55.7%	50.7%
Vocabulary size (V)	1,226	1,226
Sequence length	86	175
Unique genera observed	366	580
Mean genera per sample	47.8 ± 16.2	—

Table 8: Comparison of microbiome foundation models. ProCyon v2 achieves competitive performance with 30× fewer parameters and no pretraining, while providing per-sample LOO attribution and LLM-based interpretation.

Model	Architecture	Granularity	Pretraining	Interpretability	Weights
MGM (Ning, 2025)	8L Transformer (34M)	Genus	Next-genus, 263k	Attention + LOO	Public
BiomeGPT (Medearis et al., 2026)	Transformer + dual emb	Species	Masked abundance, 13.3k	CLS attention	Unavailable
Waypoint (Atlas/Waypoint, 2026)	Transformer (6–170M)	Genus	539k+	Not emphasized	Unavailable
ProCyon v2	SimpleEmb + MLP (1.1M)	Genus	None	LOO + LLM	Public

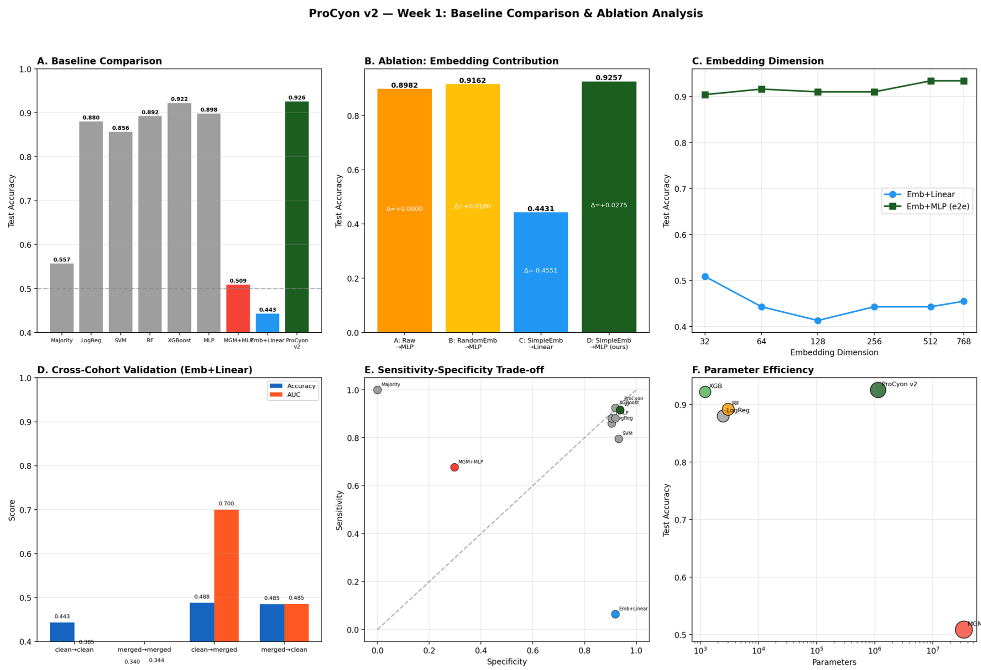

Figure 2: Comprehensive comparison analysis. (A) Baseline accuracy comparison. (B) Ablation: embedding contribution. (C) Embedding dimension sweep. (D) Cross-dataset transfer. (E) Sensitivity–specificity trade-off. (F) Parameter efficiency.

Table 9: Nested grouped-CV learning curves (AUROC). The training fraction is sampled from only the outer training groups.

Train fraction	HGB	Presence+MLP	Embedding64+MLP (ours)
20%	0.922 ± 0.027	0.940 ± 0.019	0.900 ± 0.027
40%	0.962 ± 0.012	0.959 ± 0.011	0.929 ± 0.022
60%	0.972 ± 0.010	0.965 ± 0.011	0.947 ± 0.015
80%	0.976 ± 0.009	0.966 ± 0.010	0.951 ± 0.017
100%	0.979 ± 0.010	0.969 ± 0.012	0.956 ± 0.016

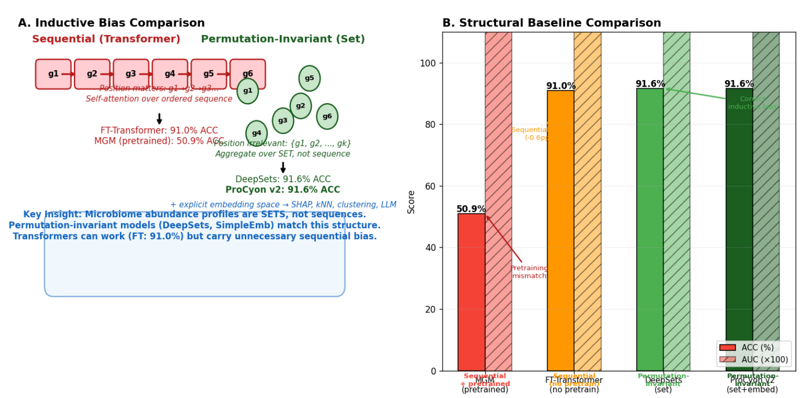

Figure 3: Inductive bias comparison. (A) Sequential (Transformer) vs. permutation-invariant (Set) inductive biases with corresponding model performance. (B) Structural baseline accuracy and AUROC comparison.

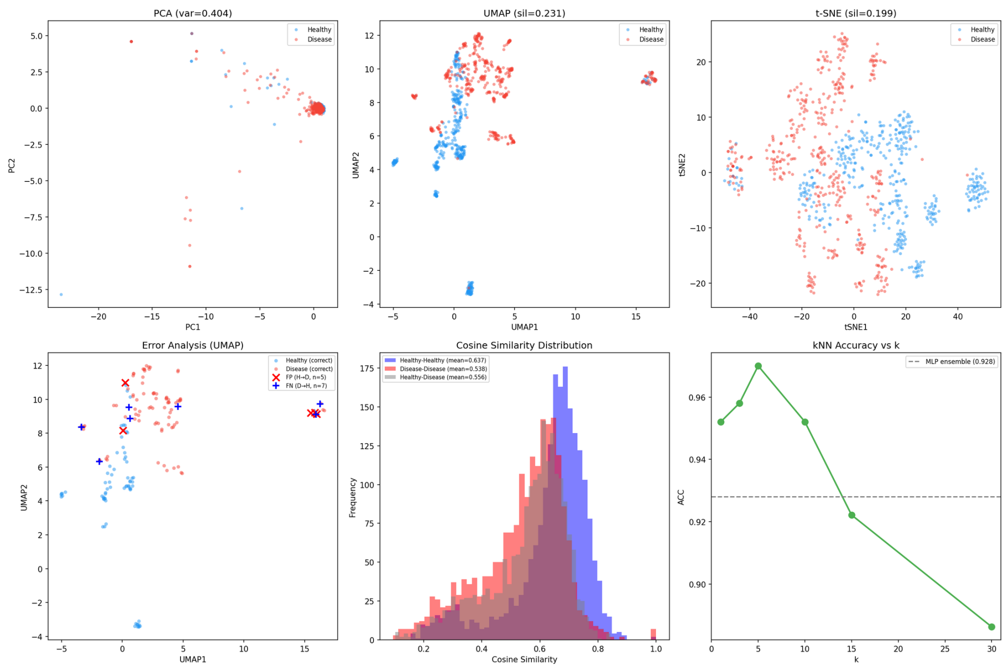

Figure 4: Representation space analysis. (A) PCA 2D projection. (B) UMAP projection (cosine metric). (C) t-SNE projection. (D) Prediction error analysis. (E) Intra-class cosine similarity. (F) kNN classification accuracy.

Table 10: LOO Attribution Reliability analysis across 5 folds.

Metric	Value
Spearman ρ (mean across fold pairs)	0.72 (p<0.001)
Top-50 Jaccard (mean)	0.07
Deletion test (top-50 AUC drop)	0.039 (vs 0.007 random)
Permutation control (mean $\lvert$LOO$\rvert$)	real 0.0096 vs. shuffled 0.0705
Literature direction agreement	5/6 (83.3%)
Prevalence-importance correlation	r=0.05 (p>0.05)

Table 11: Ablation: contribution of each architectural component. All variants use the identical MLP architecture. Δ shows improvement over variant A.

Variant	ACC	AUC	Prec.	Rec.	F1	ΔACC
A: Raw→MLP	0.8982	0.9605	0.9318	0.8817	0.9061	—
B: RandomEmb→MLP	0.9162	0.9762	0.9341	0.9140	0.9239	+1.80%
C: SimpleEmb→Linear	0.4431	0.3362	0.5000	0.0645	0.1143	−45.51%
D: SimpleEmb→MLP	0.9257	0.9750	0.9451	0.9161	0.9348	+2.75%

Table 12: Representation and head ablation under grouped-CV (15 folds). Presence = multi-hot token presence vector; Embedding64 = trainable E=64 embedding + masked mean pool.

Model	Params	ACC	AUROC	AUPRC
Presence + Linear	2,454	0.876 ± 0.037	0.949 ± 0.019	0.957 ± 0.018
Presence + MLP	315,138	0.912 ± 0.025	0.969 ± 0.012	0.973 ± 0.012
Embedding64 + Linear	78,594	0.857 ± 0.036	0.916 ± 0.029	0.909 ± 0.049
Embedding64 + MLP (ours)	96,130	0.906 ± 0.021	0.956 ± 0.016	0.954 ± 0.031

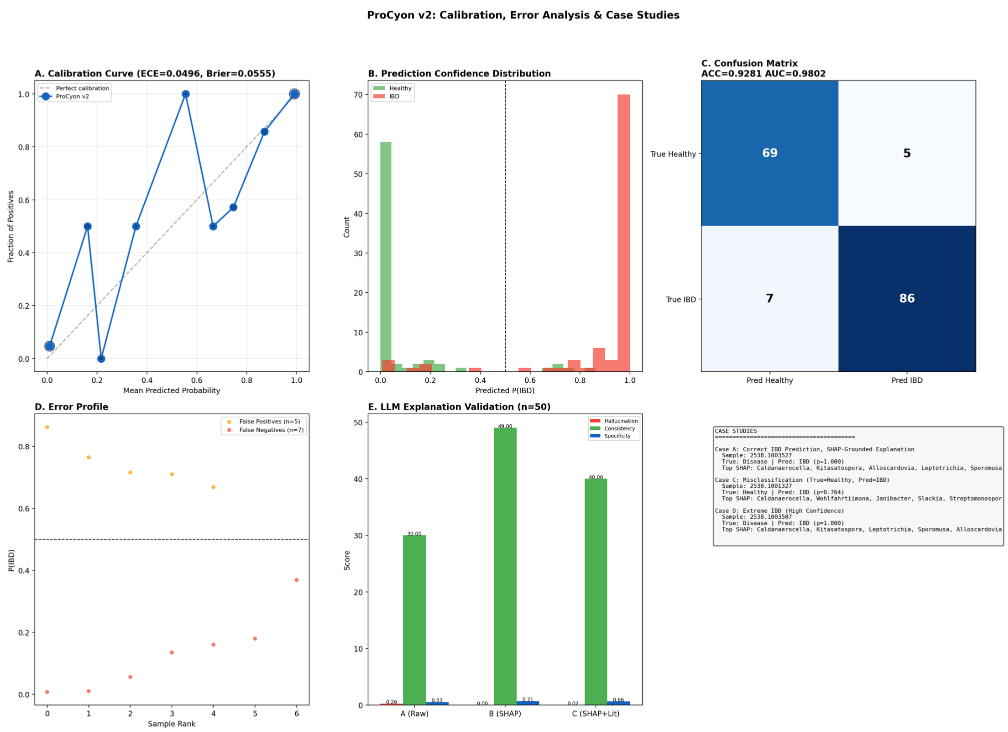

Figure 5: Calibration, error analysis, and case studies. (A) Calibration curve (ECE=0.05, Brier=0.056). (B) Prediction confidence distribution. (C) Confusion matrix. (D) Error profile: false positives vs. false negatives. (E) LLM explanation validation summary. (F) Case study overview.

Table 13: Embedding dimension ablation under the exact input-grouped 3×5-fold CV protocol. Only the embedding dimension changes; all other architecture and optimization settings are fixed. Held-out test set results are shown for reference in parentheses.

E	Params	ACC	Macro-F1	AUROC	AUPRC
16	24,994	0.855 ± 0.027	0.854 ± 0.027	0.924 ± 0.020	0.918 ± 0.041
32	48,706	0.879 ± 0.033	0.878 ± 0.034	0.941 ± 0.019	0.944 ± 0.022
64	96,130	0.906 ± 0.021	0.905 ± 0.021	0.956 ± 0.016	0.954 ± 0.031
128	190,978	0.908 ± 0.024	0.907 ± 0.024	0.957 ± 0.015	0.956 ± 0.026
256	380,674	0.909 ± 0.020	0.909 ± 0.020	0.956 ± 0.015	0.952 ± 0.037
512	760,066	0.914 ± 0.020	0.913 ± 0.019	0.957 ± 0.014	0.952 ± 0.032
768	1,139,458	0.914 ± 0.016	0.913 ± 0.016	0.957 ± 0.015	0.952 ± 0.035

Table 14: Structural baselines: inductive bias comparison. All models trained on identical data without pretraining (except MGM). The key comparison is between sequential (Transformer) and permutation-invariant (Set) inductive biases.

Model	Inductive Bias	Pretrain	ACC	AUC	Params
Raw + MLP	None (tabular)	✗	0.8982	0.9605	330K
FT-Transformer	Sequential	✗	0.9102	0.9552	769K
MGM	Sequential	✓	0.5090	0.4625	34M
DeepSets	Permutation-invariant	✗	0.9162	0.9731	512K
ProCyon v2	Permutation-invariant + Embedding	✗	0.9162	0.9640	347K

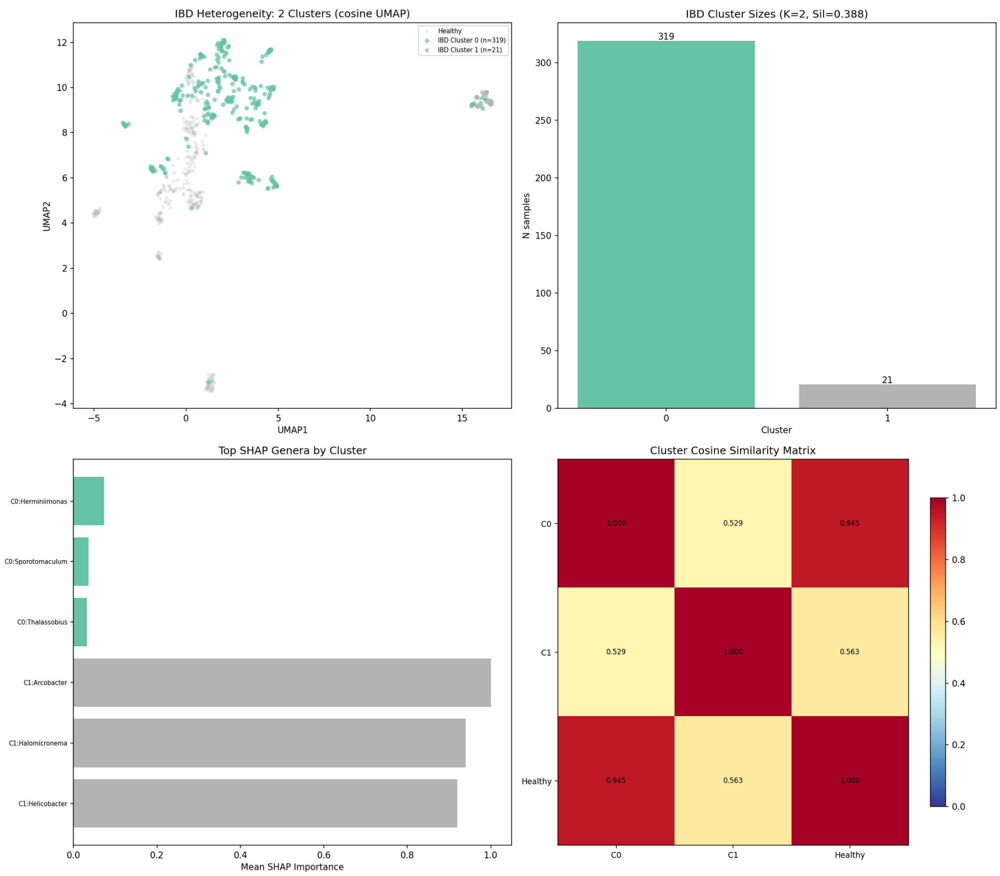

Figure 6: IBD heterogeneity analysis. (A) UMAP with IBD clusters colored separately from healthy samples. (B) Cluster sizes with silhouette score. (C) Top LOO-attributed genera per cluster. (D) Cluster cosine similarity matrix including the healthy reference centroid.

Table 15: Embedding space quality metrics.

Metric	Value
kNN (k=5, cosine) test ACC	97.0%
PCA 2D explained variance	40.4%
PCA dimensions for 90% variance	53
IBD intra-class cosine similarity	0.54
Healthy intra-class cosine similarity	0.64
UMAP silhouette score	0.23

Table 16: LOO attribution vs. differential abundance (Spearman ρ).

Prevalence Strata	ρ	p-value
All genera (n=365)	-0.23	<0.001
Prevalence $\geq$10 (n=186)	0.01	0.901
Prevalence $\geq$50 (n=108)	0.11	0.269
Prevalence $\geq$200 (n=66)	0.16	0.188

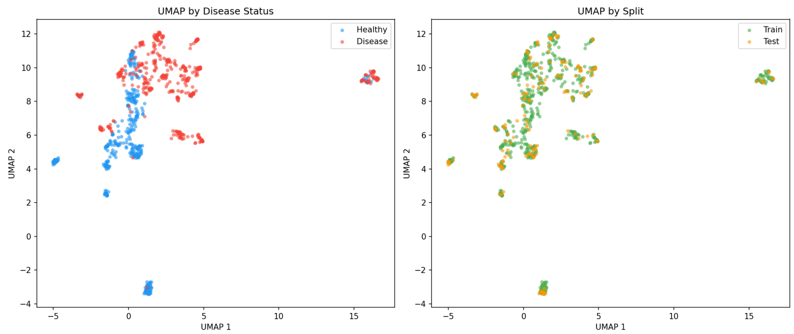

Figure 7: UMAP visualization of learned embeddings. Disease and healthy samples in the 2D UMAP projection of the 768-dimensional embedding space.

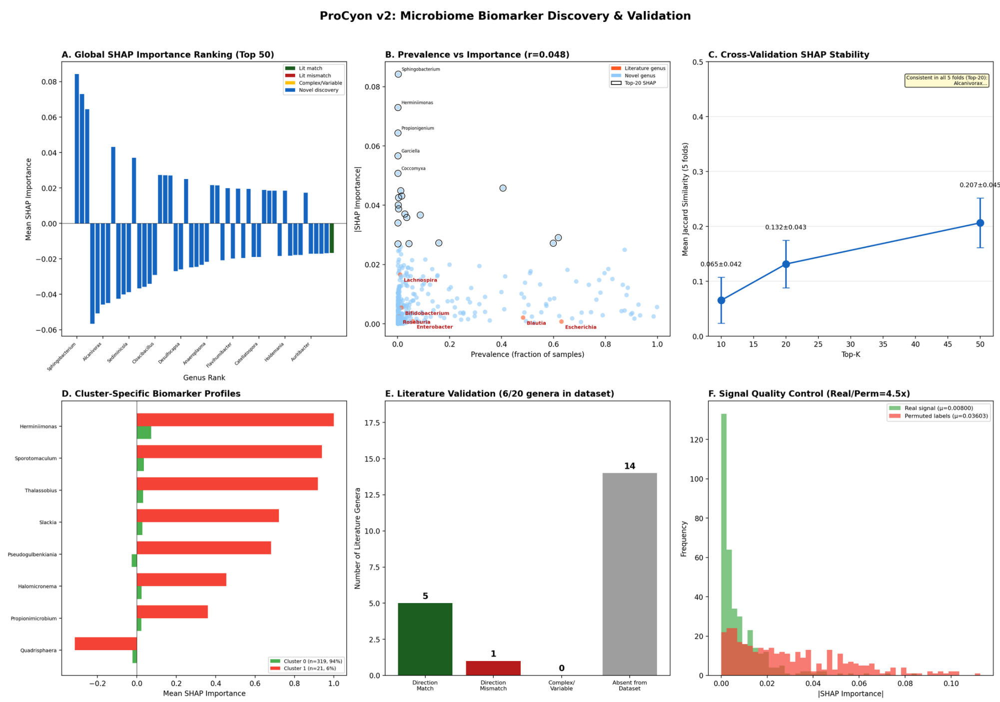

Figure 8: Comprehensive LOO attribution analysis. (A) Global importance ranking (top-50 genera, color-coded by literature agreement). (B) Prevalence vs. importance. (C) Cross-validation stability (Jaccard). (D) Cluster-specific biomarker profiles. (E) Literature direction validation. (F) Signal quality: real vs. permuted-label importance.

Table 17: Stratified LLM explanation quality (LOO+LLM variant, 167 samples).

Stratum	n	Hallucination	Consistency	Direction
Healthy samples	74	0.42	0.97	0.92
IBD samples	93	0.31	0.99	0.95
Correct predictions	155	0.37	0.98	0.94
Wrong predictions	12	0.17	1.00	0.91

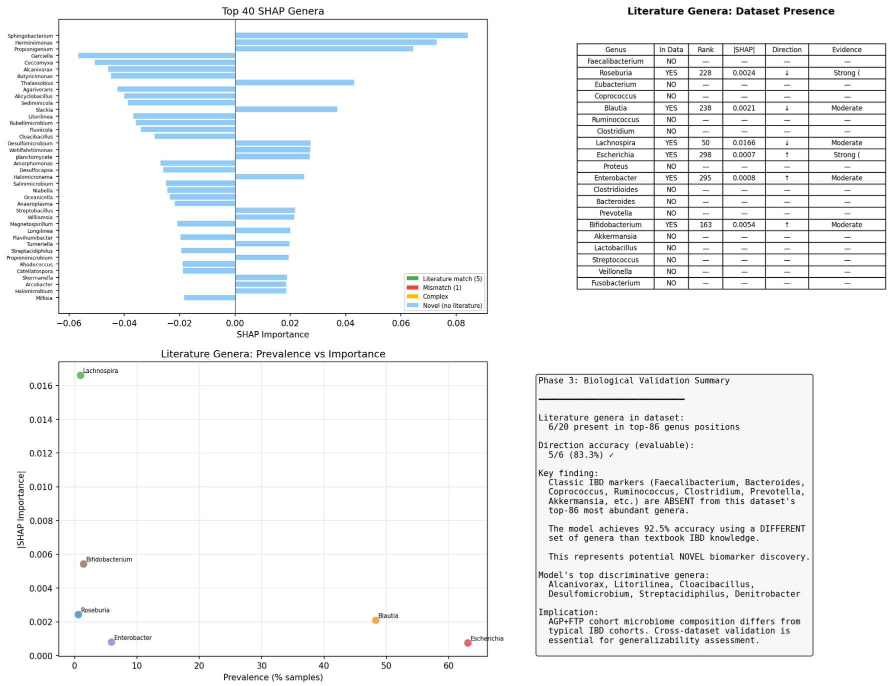

Figure 9: Biological validation against literature. Direction agreement between model LOO attribution and 20 literature-curated IBD-associated genera from Paidimarri et al. (2024).

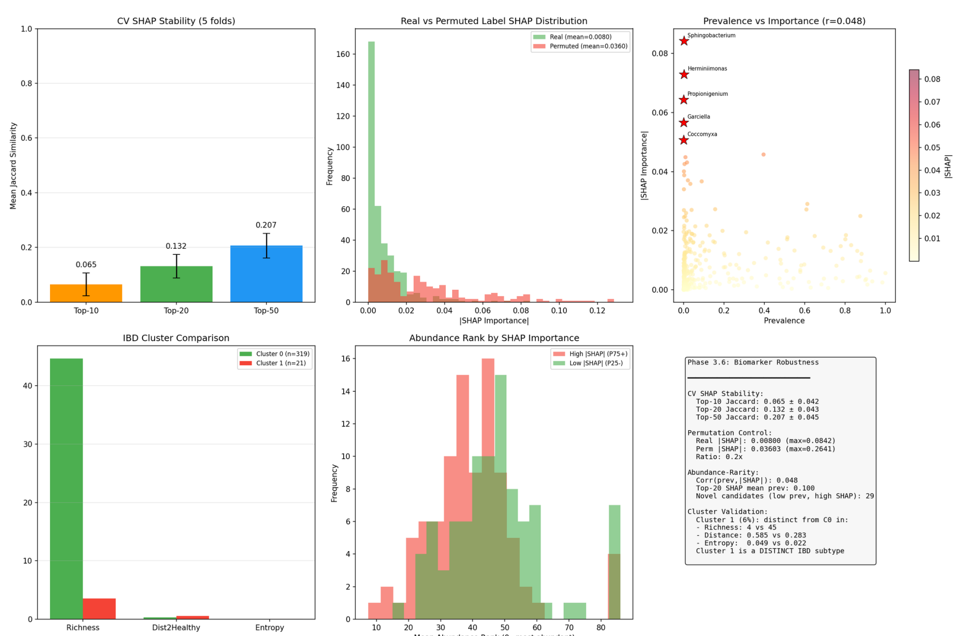

Figure 10: Biomarker robustness analysis. Cross-fold consistency, permutation control, abundance-independence, and cluster validation of LOO-attributed biomarkers.

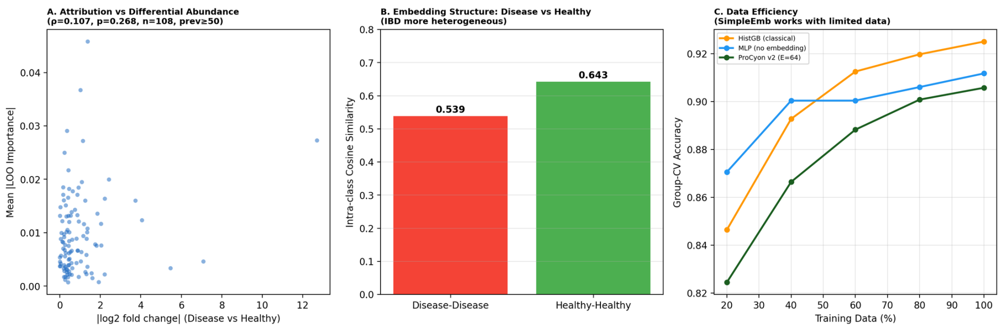

Figure 11: Attribution biology and embedding structure. (A) LOO importance vs. $|\log_2$ fold change$|$ scatter (prevalence $\geq$50). (B) Intra-class cosine similarity: disease vs. healthy. (C) Data efficiency comparison.

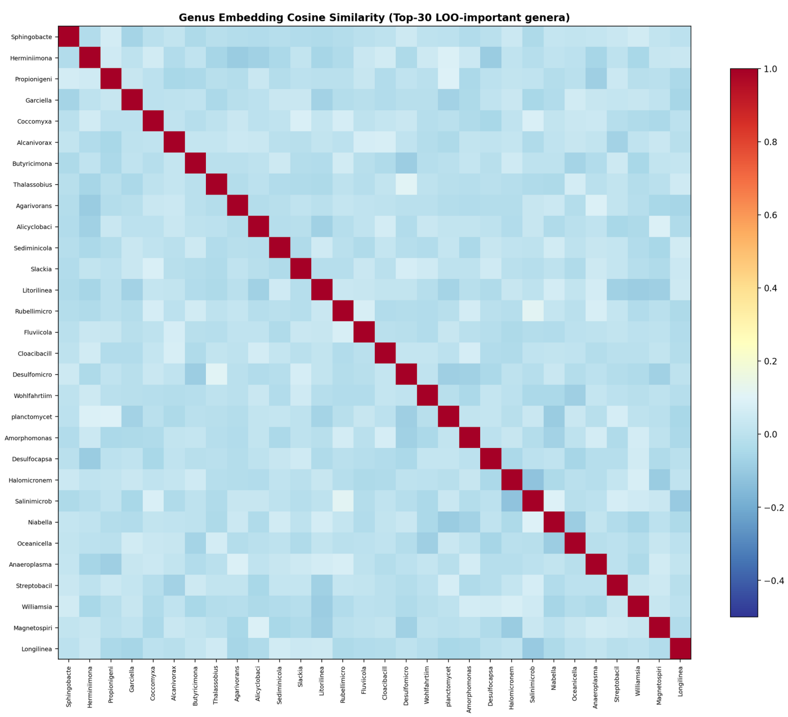

Figure 12: Genus embedding similarity heatmap. Cosine similarity among the top-30 LOO-important genera in the learned embedding space.

Table 18: Out-of-fold case analysis (grouped split, seed 42). Values are disease probabilities from three models on the same held-out samples. Extreme low-richness samples (1–4 active tokens) expose failure modes of all models.

Case	Truth	Active tokens	SimpleEmb +MLP	HGB	MGM +MLP
SimpleEmb correct; HGB wrong	Healthy	86	H (0.003)	D (0.883)	D (0.887)
HGB correct; SimpleEmb wrong	Disease	86	H (0.104)	D (0.997)	H (0.175)
SimpleEmb correct; MGM wrong	Disease	4	D (1.000)	D (0.749)	H (0.398)
High-conf. false positive	Healthy	1	D (1.000)	D (0.824)	D (1.000)
High-conf. false negative	Disease	2	H (0.000)	D (0.749)	D (0.889)
All models correct	Disease	23	D (1.000)	D (1.000)	D (1.000)

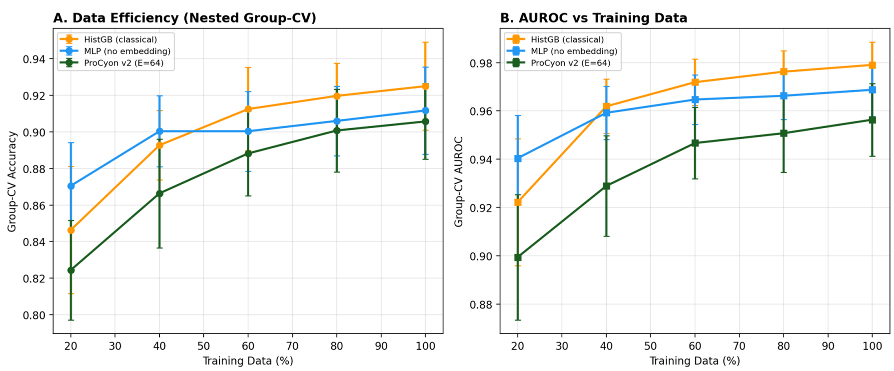

Figure 13: Data efficiency (nested group-CV). (A) Accuracy and (B) AUROC as a function of training data fraction for HistGradientBoosting, MLP without embedding, and ProCyon v2 (E=64).

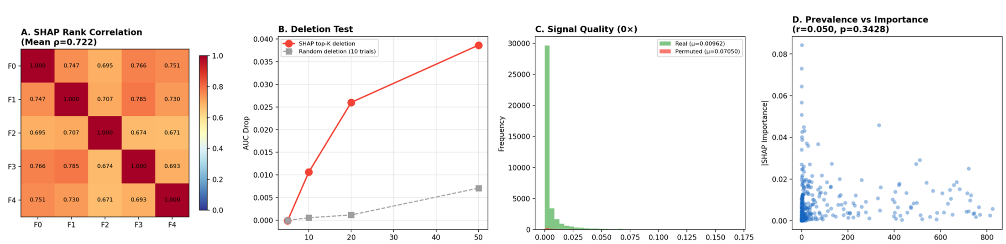

Figure 14: LOO attribution reliability. (A) Spearman rank correlation matrix across 5 folds. (B) Deletion test: AUC drop when removing top-K LOO-identified genera vs. random genera. (C) Real vs. permuted-label LOO importance distributions. (D) Prevalence vs. importance scatter.